ABHIJEET KHADE

Cloud & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Cloud & Machine Learning Engineer with hands-on experience developing, deploying, and integrating machine learning models into backend and cloud-based applications. Proficient in Python, SQL, Java, JavaScript, Scikit-Learn, Pandas, FastAPI, Flask, Django, PostgreSQL, MongoDB, SQLite, and Supabase. Experienced in building classification and regression solutions, developing REST APIs for ML model serving, processing structured datasets, and integrating trained models into application workflows. Familiar with cloud deployment, ML model deployment, application monitoring, Joblib, YOLO, rule-based systems, and fuzzy logic, with strong problem-solving and algorithmic skills.

TECHNICAL SKILLS

Programming

- **Python**
- **Java**
- **JavaScript**
- **SQL**

Machine Learning

- **Machine Learning**
- **Classification**
- **Regression**
- **Model Evaluation**
- **Data Preprocessing**
- **Feature Engineering**
- **Scikit-Learn**
- **Pandas**

- **NumPy**
- **YOLO**
- **Joblib**

Backend & APIs

- **FastAPI**
- **Flask**
- **Django**
- **REST APIs**

Cloud & Deployment

- **Cloud Application Deployment**
- **ML Model Deployment**
- **Cloud-Based ML Deployment**
- **Application Monitoring**
- **Model Serving**

Databases

- **PostgreSQL**
- **MongoDB**
- **SQLite**
- **Supabase**

Development Tools

- **Git**
- **GitHub**

Additional

- **Rule-Based Systems**
- **Fuzzy Logic**
- **Backend–ML Integration**
- **ML Pipeline Development**

PROFESSIONAL EXPERIENCE

Cloud & Machine Learning Intern

Rajyog Health and Fitness
Mumbai, India | May 2026 – July 2026

- **Developed backend services for an AI-powered Activity Monitoring and Diet Optimization platform using Python and FastAPI.**
- **Built REST APIs for integrating machine learning functionality with application workflows.**
- **Integrated a trained YOLO model into backend APIs to identify food items from images.**
- **Developed Python-based logic for calculating daily calorie intake and tracking major nutrients.**
- **Implemented machine learning and rule-based workflows for generating daily and weekly food summaries.**
- **Designed and managed application data using Supabase PostgreSQL.**
- **Created database tables and backend workflows for storing user activities and food logs.**
- **Processed structured application data using Python and Pandas.**
- **Integrated trained ML models into backend services for real-world application use.**
- **Worked on ML model deployment and model-serving workflows.**
- **Used Joblib to save and load trained machine learning models.**
- **Monitored and debugged backend and ML application workflows to improve reliability.**

PROJECTS

Cloud-Based Machine Learning Prediction Platform

Python | FastAPI | Scikit-Learn | Pandas | PostgreSQL | Joblib

- **Developed a machine learning application for predicting user outcomes using classification and regression models.**

- **Built a complete ML pipeline covering data preprocessing, feature engineering, model training, evaluation, and prediction.**
- **Used Pandas and NumPy for data preparation and transformation.**
- **Developed REST APIs using FastAPI to serve trained machine learning models.**
- **Saved and loaded trained Scikit-Learn models using Joblib.**
- **Integrated PostgreSQL for storing application and prediction data.**
- **Designed backend workflows for submitting data and receiving ML predictions through APIs.**
- **Prepared the application for cloud-based deployment and ML model serving.**
- **Implemented model evaluation workflows to compare prediction performance.**

AI-Based Image Classification & Analysis System

Python | YOLO | FastAPI | PostgreSQL | Machine Learning

- **Developed an AI-powered image analysis application using a trained YOLO model.**
- **Integrated the trained model into a FastAPI backend for real-time image processing.**
- **Created REST APIs for uploading images and returning model predictions.**
- **Implemented backend workflows for storing prediction results and related application data.**
- **Used PostgreSQL for structured data storage.**
- **Integrated ML inference into a production-oriented backend application.**
- **Designed the system to support deployment of the trained model through a backend API.**

Fitness Prediction & Recommendation System

Python | Django | Scikit-Learn | Joblib | Machine Learning

- **Developed a machine learning-based application for analyzing user fitness and health-related data.**

- **Implemented classification and regression workflows for generating predictions.**
- **Used Scikit-Learn for model development and evaluation.**
- **Applied preprocessing and feature engineering to prepare user data for ML models.**
- **Used Joblib to save and load trained models.**
- **Developed Python backend scripts for automated prediction and recommendation workflows.**
- **Implemented rule-based and fuzzy-logic functionality alongside machine learning models.**
- **Generated automated weekly plans using backend Python workflows.**

CLOUD & ML ENGINEERING COMPETENCIES

Machine Learning Lifecycle

Data

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Data Preprocessing

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Feature Engineering

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Model Training

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Model Evaluation

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Model Serialization

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REST API

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Cloud Deployment

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Prediction / Monitoring

Core Areas

- **Machine Learning Model Development**
- **Classification**
- **Regression**
- **Data Processing**
- **Feature Engineering**
- **ML Model Evaluation**
- **ML Model Serving**
- **REST API Development**
- **Backend Integration**
- **Cloud Deployment**
- **Database Integration**
- **Application Monitoring**

EDUCATION

Vidyalankar Institute of Technology — Mumbai

Bachelor of Technology in Computer Engineering
2023 – 2027

CORE STRENGTHS

- **Machine Learning Engineering**
- **Python Development**
- **Cloud-Based ML Applications**
- **ML Model Deployment**
- **REST API Development**
- **Backend Development**

- **Database Management**
- **Data Processing**
- **Classification & Regression**
- **Algorithmic Problem Solving**
- **Git & GitHub**
- **Software Engineering**